

Additive batch electrospinning patterning of tethered gelatin hydrogel fibres with swelling-induced fibre curling

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ABSTRACT

Limited methods exist to integrate patterning of free-spanning hydrogel microfibres as part of an additive manufacturing strategy. We demonstrate 3D additive batch electrospinning (3D-abES) workflow to efficiently produce layered hydrogel gelatin microfibres, tethered to 3D printed thermoplastic structures of various shapes. Enabled by the digital-design approach, multiple 3D printed fibre devices can be fabricated in a batch efficiently with minimised sample-to-sample variance. Customised fibre patterns and device geometries can be rapidly altered to fit with potential applications in 6- to 24- well plate formats. The diameter of as-produced dry fibres is in a range of $\sim 2-4 \mu\text{m}$. With the intended device applications in aqueous environments, we investigate the effect of combinations of processing parameters on the gelatin fibre integrity, and its swelling behaviour when immersed in water. Aided by the parametric study, patterns of swelling-induced fibre curling, from straight to wavy, can be tuned. Our findings could serve as a guide to optimise the 3D fabrication and patterning of electrospun hydrogel-like fibres made of crosslinked hydrophilic polymers and oligomers, for cell culture or bio-sensing applications.

1. Introduction

Electrospinning techniques offer a myriad of design options for micro-to-nano scaled fibre production, with a broad range of functionalities [1–3]. Intertwining 3D printing with electrospinning technologies opens up prospects in producing new architectures not possible with a single material deposition technique, demonstrating novel applications ranging from hydrogel actuators to 3D tissue engineering scaffolds [4–8]. Electrospun fibres can be produced at a similar scale length to those in the extracellular matrix of tissues, subsequently biocompatible materials have been extensively used to build *in vitro* cell culture models, demonstrating promising potential in tissue engineering applications [9–11]. In particular, with assistance from 3D printed substrates, 3D electrospun fibrous scaffolds have been shown as an effective strategy to realise 3D cell culture [12–14]. Although existing studies have demonstrated inter-changeable processes between 3D printing and electrospinning [8,15,16], controllable, rapid deposition and assembly of electrospun fibres in 3D architectures remain technologically challenging to produce in an integrated manufacturing scheme. When operating as the sole material deposition technique, electrospun fibre patterning is capable of producing randomly

organised mat structures, to highly ordered 2D and designable 3D patterns [17–19]. Fibrous scaffold architecture in crimped and wavy conformations have been further demonstrated [20–25]. These existing fibre patterning strategies, however, could be difficult to implement in conjunction with 3D printing, due to the drastically different fabrication mechanisms, mechanical properties and feature sizes of the products resultant from electrospinning *versus* 3D printing. Electrospinning relies on an electrical static force to draw high aspect ratio, flexible fibres in the micro- to nano- scale; while commonly used 3D printing techniques produce parts that are mechanically rigid and have sub-millimetre print resolution. Therefore, integrating both mechanisms in a compact and automated platform to realise rapid and stable fabrication of 3D fibre scaffolds remains a challenge for both process design and material-property optimisation.

Furthermore, most existing studies have primarily focused on the electrospinning patterning strategies of thermoplastic fibres due to their intended applications as bio-resorbable implants [17,26]. In contrast, the patterning strategies for electrospun hydrogel fibres consisting of amino acid based, and/or crosslinked hydrophilic materials, are less explored. Compared to synthetic thermoplastic polymers, gelatin is a natural biomaterial derived from collagen, providing better mimicry of

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many features of the native extracellular matrix environments [27,28]. Gelatin forms an electrospinnable solution with a mild solvent system, whilst retaining many of the desirable properties of collagen. These properties include inherent arginine-glycine-aspartic acid (RGD) peptide sequences for cell integrin attachment [29], and cleavage sites for MMP-2 and MMP-9 [30]. In addition, gelatin hydrogels have been shown to exhibit tuneable mechanical properties matching those of the natural soft tissues through varying crosslinking density [31,32]. The combination of desirable biocompatibility, biodegradability and mechanical properties make gelatin a favourable material for *in vitro* ECM modelling. Extending from gelatine hydrogels, fibrous hydrogel materials have also gained general interests in potential applications such as tissue engineering scaffolds, *in vitro* models for studying cell-matrix interaction, and environmental sensors [9,33–38].

We recently demonstrated a new 3D low-voltage electrospinning patterning (3D-LEP) technique [39], which integrates solution-based low-voltage electrospinning (LEP [19]) with fused filament fabrication (FFF [40]). 3D-LEP produces aligned, suspended micro- and nano-fibres tethered to 3D printed thermoplastic supporting structures. By using crosslinked gelatin as an example material, the hydrated gelatin microfibres exhibit a Young's modulus that resembles soft-tissues [41,42]. For ongoing utilisation of such fibre devices for a broad range of biological and analytical assays, it is of significant interest to parallelise production and explore the versatility of the hydrated fibre patterns beyond the previously demonstrated linear, straight-line conformation [41].

Herein, we demonstrate a 3D additive batch electrospinning (3D-abES) workflow, that rapidly fabricates multi-layered 3D fibre devices in an automated process. By optimising the toolpath and digital-design approach, multiple devices can be efficiently produced as a batch with minimised sample-to-sample variance. We show that the 3D-abES workflow allows facile adjustment of the device configuration and enables convenient, low-cost and efficient batch production to fit with potentially versatile end-use applications [43]. Since the devices were developed for applications in aqueous environments, the processing parameters that influence fibre curling in water were investigated. As gelatin is a hydrophilic polymer and naturally-derived protein, our findings may also serve as an important guide to optimise the fabrication and utilisation of generic hydrogel-like microfibres for various biological and chemical assay developments.

2. Materials and methods

2.1. System design

The operation of 3D-LEP was adapted to establish the automated 3D additive batch electrospinning (3D-abES) workflow. As shown in Fig. 1a–d, the fabrication process involves switching between two material deposition techniques, namely the fused filament fabrication (FFF) for thermoplastic (Polylactic acid, PLA) printing (Fig. 1a–a₁), and the low-voltage electrospinning (LEP) process for producing gelatin fibres tethered to the PLA structures (Fig. 1b–b₁). The two printing heads can be automatically exchanged during the printing process thanks to a magnetic tool-mount, as shown in Fig. 1c. The FFF process prints a support structure, on top of which the microfibres are patterned. In addition to acting as a support structure of the microfibres, the PLA frame also acts as an 'initiator' to break up the electrospinning solution droplet for fibre initiation (Fig. 1d). Since 3D-abES produces dry fibres with a fixed spanning length defined by the distance between the pair of PLA supporting pillars, hydrogel fibres curl as a result of increased fibre volume due to solution uptake (*i.e.* swelling) compared to the dry fibre states (schematically shown in Fig. 2). Hence, the originally organised fibre pattern deposited in the dry state can be altered to different extents in a controlled manner. By adjusting the fibre drawing speed, crosslinker concentration and crosslinking time, the resulted fibre morphologies could be designed to possess various levels of curling.

Despite of fibre swelling upon hydration, tethering the fibres to a rigid thermoplastic structure preserves the structural integrity of the entire 3D fibre device.

2.2. Instrumentation

The firmware of an Ultimaker 2 (Ultimaker, the Netherlands) 3D printer was modified in-house to recognise a secondary printing tool, and a command to switch between the tool-heads. For FFF printing, the printer was used as it was commercially designed (Fig. S1.a). For the electrospinning process, the printer provided robotic X-Y-Z translation (Fig. S1.b). The design of both printheads are shown in Fig. S1.a₁ & b₁, and the design of the docking stations of both printheads was derived from STL models, which are available on an open-source 3D printing file deposit: <https://www.youmagine.com/designs/dual-head-ultimaker2>. This design was modified to enhance the patterning and printhead exchange precision utilising kinematic coupling to improve the mounting tool stability. In order to control the solution flow rate for the electrospinning process, additional apparatus consists of a pneumatic pressure controller (OB1 MK3 system, Elveflow, France) supplied with a nitrogen gas canister. The gelatin solution was constantly fed under a 30 mbar air pressure during the whole printing process. To create the static electric field, a digital DC voltage source (Stanford Research PS350) was connected to a 1 mL stainless steel syringe adapter (AD931-6450, Advanced Adhesive Solutions Ltd) and a rectangular brass grounding plate designed to fit on top of the printer's glass build plate. A syringe barrel (1 cc 401 L-1 series, Advanced Adhesive Solutions Ltd) was connected to the pressure source *via* a barrel adapter assembly (900-250-6 Advanced Adhesive Solutions Ltd), and the syringe adapter has a blunt metallic syringe tip at the opposite outlet (25 gauge, TE725050PK). During the LEP process of the 1st and 2nd fibre layer, the distance between the tip of the needle and the grounded plate is around 1 mm and 1.3 mm respectively, and the voltage was maintained at around 100 V and 130 V respectively. Generally, when building on height, the applied voltage should increase with the distance between the tip of the needle and the grounded plate, and the overall static field should be kept at around 100 V.mm⁻¹. Tool head movement speed (or fibre drawing speed) was set between 15,000 and 9000 mm/min. Experiments were performed under humidity of 60–80 %.

2.3. Software configuration

The firmware on the Ultimaker 2 3D printer was modified *via* Arduino IDE (Arduino) through the Marlin open-source repository on Github (github.com/Ultimaker/Ultimaker2Marlin). This allowed the printer to identify where to pick up and replace tool-heads for the FFF and the electrospinning processes and permitted calibration of the co-ordinate system off-sets. CAD models were designed and converted to STL files in Autodesk Inventor Professional 2017 (Autodesk Inc.). 'Simplify 3D' slicer software was used to generate G-Code. The G-Code was then manual modified in Notepad ++ where necessary adjust tool-path movements. The electrospinning solution flowrate was modulated through the pressure supplied to the syringe barrel from an Elveflow microfluidic flow control system, directed by the proprietary Elveflow ESI software (Elveflow, France).

2.4. Fibre materials and crosslinking

A gelatin solution with a concentration of 19 % w/w was prepared from porcine gelatin (Porcine skin, Sigma-Aldrich) dissolved in a previously optimised solvent composition of: acetic acid (33 % w/w), ethyl acetate (22 % w/w) and de-ionised water (26 % w/w). The solution was stirred for 24 h at room temperature [44,45]. Here, glyoxal was selected as the crosslinking agent as it has similar reaction characteristics to formaldehyde and glutaraldehyde but has a lower toxicity level [44]. Its

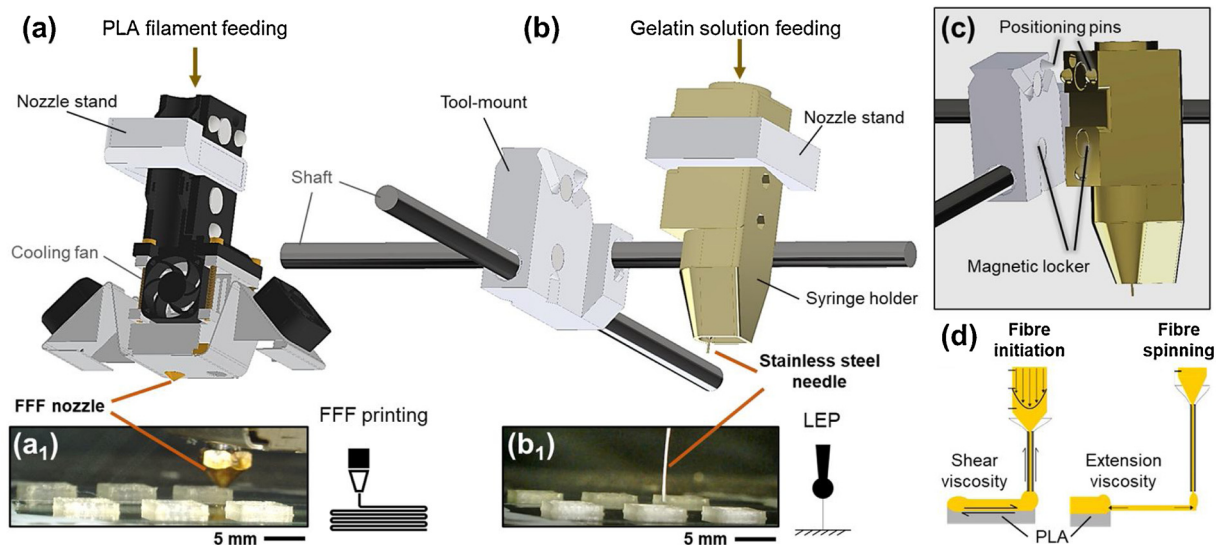


Fig. 1. The additive batch electrospinning (3D-abES) workflow consists of (a-a₁) the fused filament fabrication (FFF) and (b-b₁) the low voltage electrospinning (LEP) process. (c) Scheme showing the switching mechanism between the FFF and LEP processes. (d) Scheme showing the plastic PLA structure acting as an ‘initiator’ for breaking up the electrospinning solution droplet for fibre initiation, as well as the structure for fibre tethering.

crosslinking mechanism is homobifunctional. Glyoxal of different concentrations (between 3–10 % v/v) was added to the gelatin solution as a chemical crosslinker prior to electrospinning. After the fibre devices were fabricated by additive batch electrospinning, they were left overnight at room temperatures to ensure that the crosslinking reaction was completed [44].

2.5. Fibre characterisations

Images of fibre structures were taken using a combination of a Leica optical microscope (Leica DMLM + EC3 camera, Leica Microsystems) and a GX Capture microscope camera (GT Vision Ltd.). The fibre diameter measurements were taken from scanning electron microscope (Helios Nanolab 650 DualBeam, FEI) images. Samples were affixed to metal SEM mounts with double-sided conductive carbon tape. Images were acquired using an accelerating voltage of 500–700 V, and this voltage was low enough to avoid the need to sputter coat them with gold, which might damage the suspended fibres. To image the global morphology of fibre arrays, magnifications within the range of x 133–500 were selected. To measure individual fibre diameter, a magnification of x30,000 was chosen and 171 fibres were imaged overall. Crosslinked dry fibres were placed in DI water for over 12 h

prior to morphological assessment of the swollen state. Although hydrogel swelling is typically measured by a swelling ratio comparing wet versus dry weight, this would be impractical given the scale of the fibre features. Accurate quantification of the hydrogel diameter and hydrogel volume post swelling is also challenging as the fibres are ‘free floating’ in 3D over millimetre distance range. To overcome these challenges, we adapt swelling-induced fibre elongation, as an indication of the lower limit of fibre swelling ratio. Swelling-induced fibre elongation is defined as the change in fibre length before (L_0) and after liquid soaking (L_s), with elongation = $(L_s - L_0) / L_0$, where both L_0 and L_s are projected fibre lengths (thus a lower limit of elongation was estimated).

3. Results and discussion

3D-abES operates based on a layer-by-layer fabrication strategy. In order to facilitate the 3D-abES production workflow (schematically illustrated in Fig. 3), the PLA structures for all the designated devices were printed all at once as the first layer (*i.e.* 1st PLA layer) with FFF process (Fig. 3a). Subsequently, the print head is switched to the electrospinning head for fibre array tethering (as the 1st fibre layer) on the existing PLA supporting structures (Fig. 3b). A voltage source applied on the needle provides an electric field between the droplet and

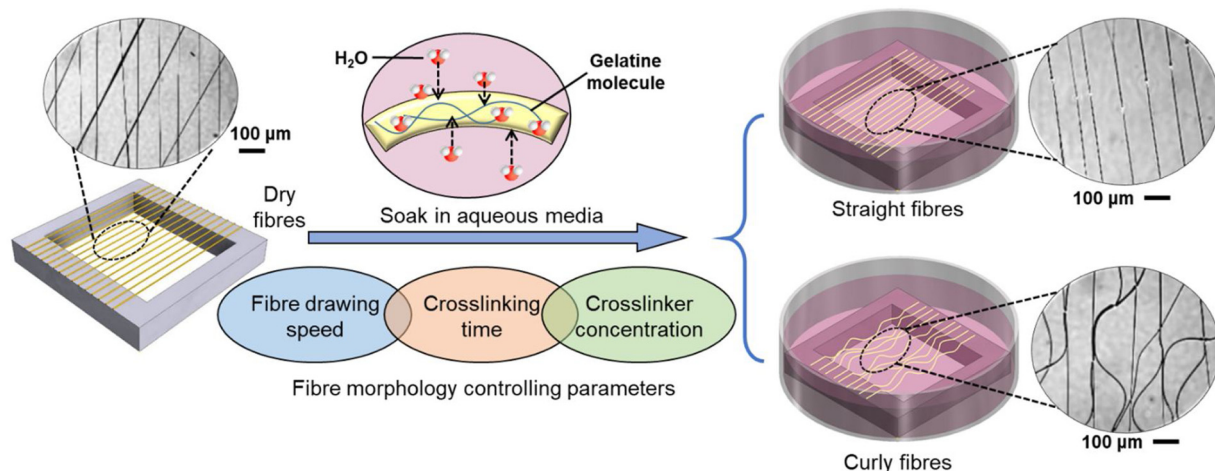


Fig. 2. Scheme showing the proposed parameters controlling the swelling-induced fibre curling behaviour.

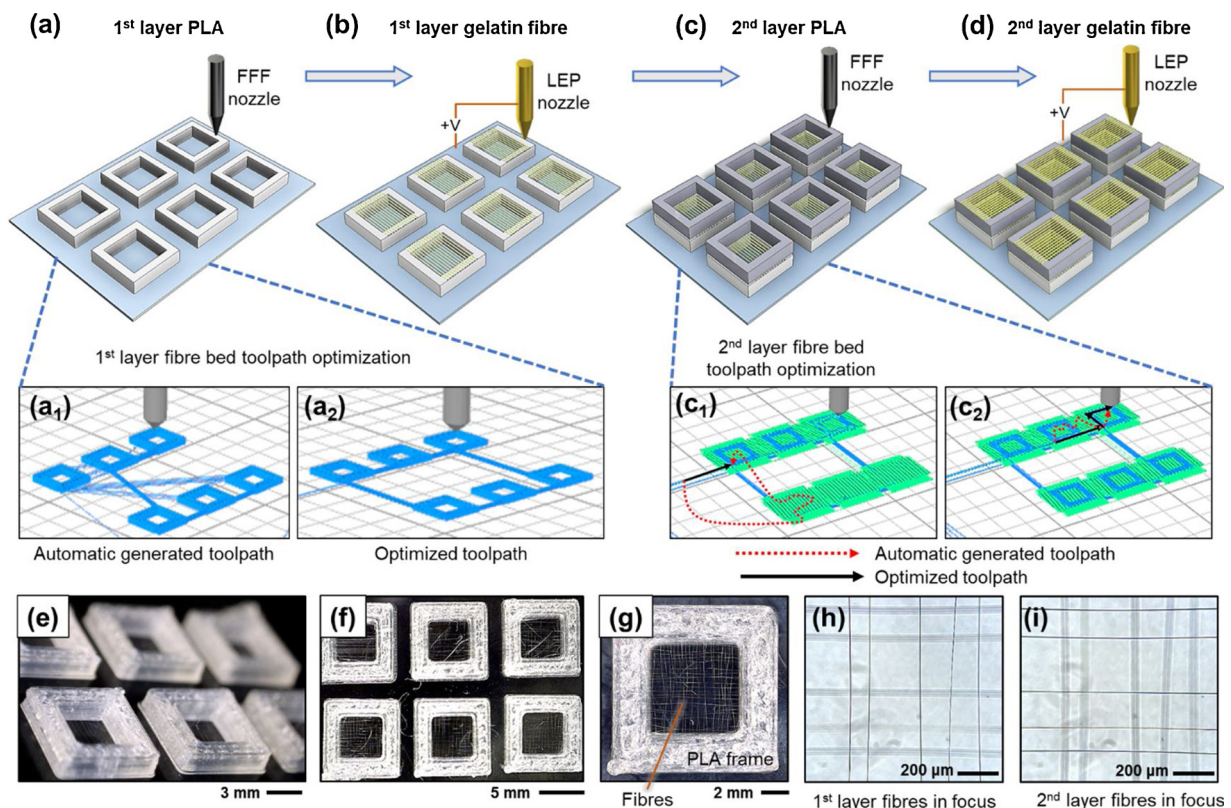


Fig. 3. Tool path optimisation in a layer by layer fashion. (a) First PLA layer printed by the FFF nozzle, with schemes showing (a₁) the tool path generated by default settings, versus (a₂) the optimised tool path. (b) First fibre layer printed by the electrospinning nozzle. (c) Second layer PLA printing, with schemes comparing two process designs in (c₁–c₂). (d) Second fibre layer printed by the electrospinning nozzle. (e–f) An example of six fibre devices produced in the same batch, showing images captured at different views. (g) One of the double-layer fibre device, with microscope images focusing on the first (h) and second (i) fibre layer.

the grounded build plate. This tensile force helps stretch the droplet and thin the fibre. The process repeats, forming three-dimensional layered fibre structures, as shown in Fig. 3c–d. An overview of the printing sequence is shown in Supplementary Movie-1. In the previous studies of 3D electrospun fibrous structures, although FFF printed plastics have also been used as substrates for fibre deposition, it is not straightforward to build multi-layered structures with highly-ordered microfibres [14,46]. This could be mainly because in these approaches, the FFF printing and electrospinning are conducted separately, and manual intervention is involved during the process. In the 3D-abES system, the automated printhead switching mechanism offers the versatility of constructing multi-layered fibre structures in height and the efficiency to mass produce samples in a batch.

It is found that three broad considerations could determine the quality of the fibre device: First, disruption to fibre placement due to the defects of the PLA structures; second, fibre damage caused by the interference of toolpath with the as-spun fibres; third, different extents of fibre damage when the crosslinked fibre device is immersed in an aqueous solution. In the following, we will discuss that how the fine-tuning of the printing toolpath design could minimise the print defects, leading to controllable and uniform high-quality devices. On the basis that the electrospinning solutions already possess certain characteristics for the fibre production, we further investigate how combinations of fabrication parameters (i.e., Fig. 2) can affect the batch-processed fibre quality leading to controllable fibre curling behaviour in solution.

3.1. Adapting PLA layer printing as supporting structures

For batch printing of several PLA structures in parallel, the FFF print head is required to have more travel movements in between objects than printing a single object. Thus, the print files are found to be more

susceptible to suffer from over- or under-extrusion of the thermoplastic filament due to frequent extrusion and retraction of the filament. First of all, over-extrusion leads to oozing and stringing defects that can interfere with subsequent fibre placement. Tuning the retraction settings in the PLA printing can reduce this problem: this refers to the act of reversing the filament into the print head when performing travel movement to halt filament extrusion. However, the comparatively frequent, repetitive short travel movements during printing multiple PLA structures in a batch (Fig. 3a₁) could cumulate overtime to result in under-extrusion. Under-extrusion of the PLA support structure is also highly detrimental. At an early stage of a print, this leads to poor build-plate adhesion and potential detachment of an unfinished device. At a later stage, this can lead to ‘stringing’ defects and a rough initiating surface for fibre patterning. Surface roughness could lead to chaotic and variable fibre initiation. The G-Code print files were manually adjusted to detour around the outline of a printed object, minimising and isolating travel movements to specific regions, as shown in Fig. 3a₂. This was an effective strategy to avoid PLA extrusion defects forming in batch prints of suspended fibre devices.

3.2. Building in height: subsequent layer printing of PLA and fibre arrays

For building a three-dimensional fibre array device, multiple layers of PLA supported fibre arrays should be printed in a stacked manner. After completing one round of PLA-fibre patterning, the LEP nozzle is retrieved to exchange to FFF printhead. During this process, the automatically generated default tool exchange sequence produces interfering travel movements of the hot-ended FFF printhead translating directly across as-patterned gelatin fibres (Fig. 3c₁). These interfering print paths cause damage to the fibres already patterned in the previous round. Either editing the specific command lines in the G-Codes or

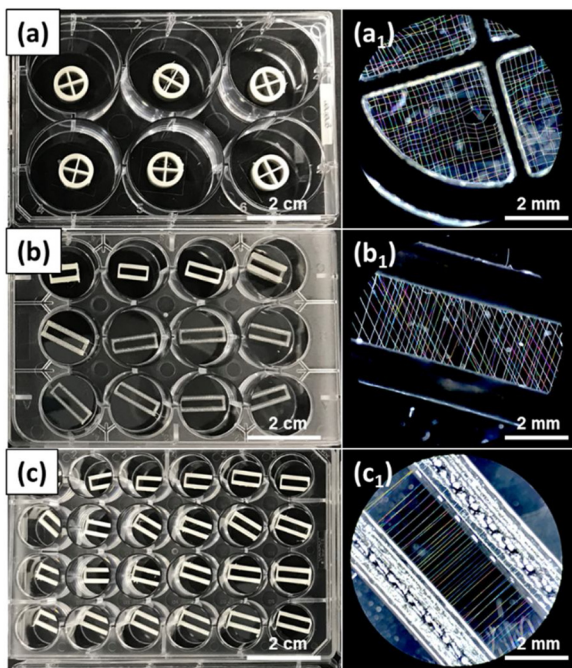


Fig. 4. Designable fibre devices for use in (a) 6 well-plates; (b) 12 well-plates; and (c) 24 well plates. Zoomed-in images are shown for each fibre pattern: (a₁) double-layer with 90° off-set, (b₁) double-layer with 45° off-set, and (c₁) single layer with aligned fibres.

adjusting the tool exchanging sequence of commands as shown in the preview of Fig. 3c₂ could solve this issue. The software settings were modified so that the toolpath of the printhead avoids crossing the outline of any individual printed parts. A successful batch print of devices enables parallel and automated fabrication of multi-layered 3D fibre devices, which improves the production efficiency. In addition, this also optimises the utilisation of the electrospun solution by reducing the working time span.

In less than eight minutes, six double-layered gelatin microfibre devices (each of 7 mm × 7 mm × 1.5 mm) could be printed. Fig. 3e–f shows macroscopic views of the six example fibre devices, each with a 90° off-set placement between the first and the second fibre layers, and Fig. 3g shows an individual device with the double-layer gelatin fibres. Zooming-in at the fibre layers in and out-of-focus clearly show the fibres were orderly patterned (Fig. 3h–i); and that the two orthogonally-oriented fibres were suspended at two focal planes, thus at different vertical positions. Based on the principles of digital-design to print-path setup stated previously, arbitrary PLA fibre support structures, fibre patterns and alignment off-sets can be designed depending on the end-use applications. As shown in Fig. 4, the batch processing platform enables scaled-up fabrication of different device types to be fitted in 6 well (Fig. 4a), 12 well (Fig. 4b) and 24 well (Fig. 4c) plates that are commonly used in cell culture and biochemical assay applications. The

corresponding CAD models of each PLA structure design have been included as Figs. S2–S4 respectively. Such production ability is foreseen to be useful for performing experimental replicates.

3.3. Studies for controlled fibre properties

Sample-to-sample variance is an essential quality control characteristic of the batch manufacturing process. Maintaining a similar fibre diameter among the devices fabricated in one batch could benefit potential applications. Printing devices as a batch allows independent assessment of both the fibre drawing speed and the effect of cross-linking time as separate factors on the fibre diameter distribution among the samples. A range of increasing tool translation speeds (as a means of increasing the rate of fibre drawing) were investigated for different crosslinking times, with the crosslinking concentration fixed at 6 v/v %. For each experimental condition, diameters of dry fibres were measured using bright-field microscopy. Fig. 5a shows that the dry fibre diameter is uniform when printed within the 3000–6000 mm/min fibre drawing speed and there is no clear correlation between the fibre size and crosslinking times in this speed range. This is because that in the LEP process, the fibres are formed mainly by the tensile force exerted between the PLA frame and the needle. Thus, the different fibre drawing speeds might not affect the tensile force and fibre size drastically because the as-extruded fibre jet remains a semi-liquid state before the solvent evaporation. In addition, the extruded solution immediately at the tip of the needle is always in excess compared with the solutions that form the fibres at all fibre drawing speeds discussed. Beyond a drawing speed of 6000–7500 mm/min, the fibre size becomes less uniform, especially later in the crosslinking reaction as fibres had a greater tendency to merge as initiation lost regularity. It became difficult to increase the flow rate to counteract the increasing solution viscosity.

Based on the intermediate drawing speed of 4500 mm/min, we assessed the electrospinning reproducibility. The dry fibre diameter of uncrosslinked fibres was measured using SEM, providing an indication of reproducibility. As shown in Fig. 5b, the gelatin microfibre, threaded onto the PLA supporting structures, are uniformly suspended in parallel array pattern. Zoomed-in image in Fig. 5c shows the single fibre morphology. The gelatin microfibre is around 2 μm thick, with a smooth surface texture. Across six devices the mean fibre diameters across the different samples all fall within 2–4 μm (Fig. 5d). The lower mean values measured by SEM compared to those found by bright-field microscopy (~3–5 μm) is reasonable due to the difference in electromagnetic wave diffraction limits of the two imaging techniques. Nonetheless, the upper fibre diameters, which should be subjected to less diffraction limit constraints, were reported to be consistently ~8 μm for both techniques. The relatively large variation in the fibre diameter range is possibly due to the pressure-driven flow rate control of the pneumatic system.

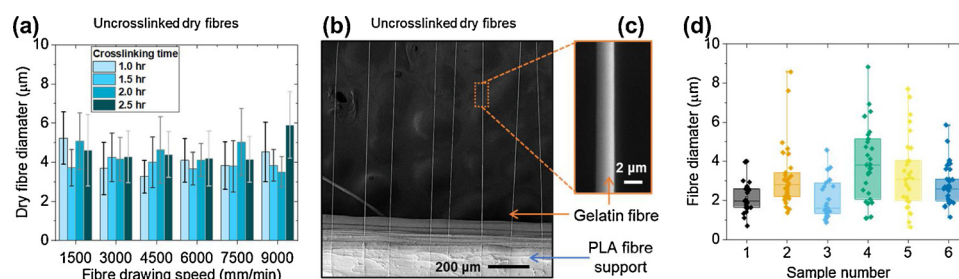


Fig. 5. (a) Histogram showing dry fibre diameter patterned at different crosslinking times and fibre printing speeds. (b–c) Environmental scanning electron microscopy of dry uncrosslinked fibres. (d) Variation of fibre diameters across 6 different samples of devices.

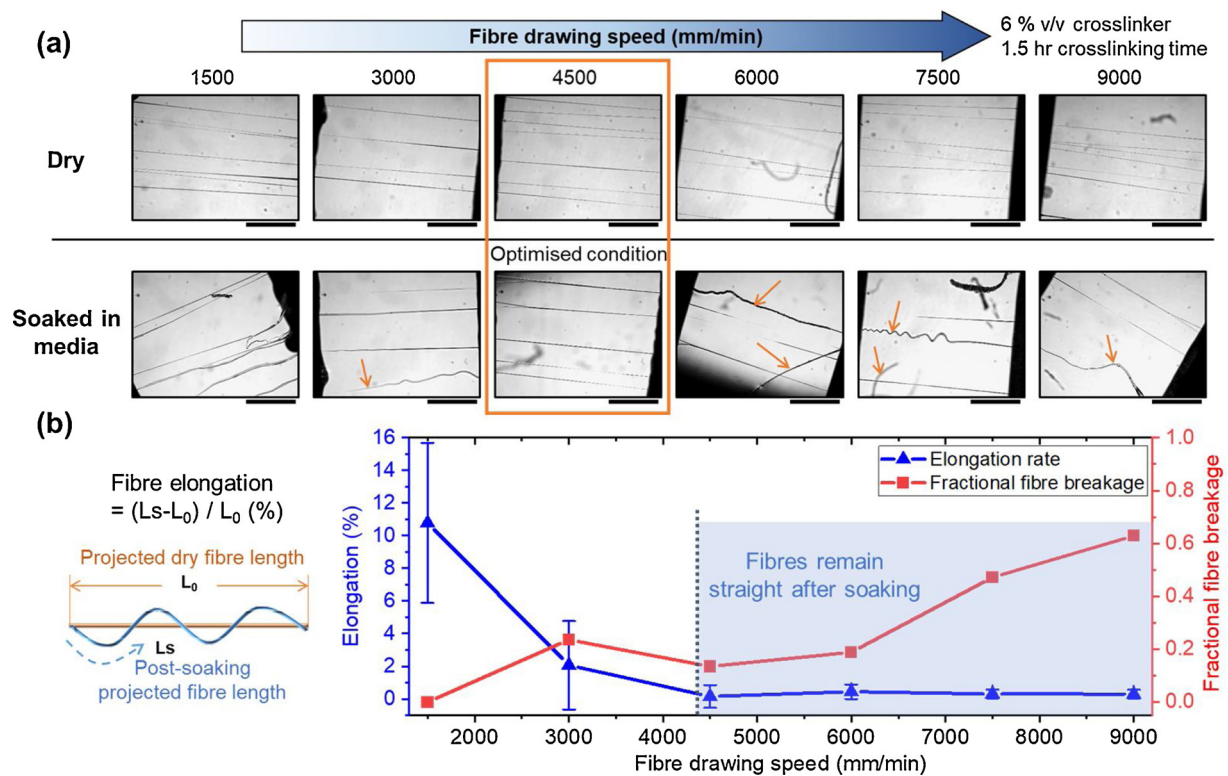


Fig. 6. (a) Images showing gelatin fibres fabricated at different fibre drawing speed (scale bar: 150 μm). (b) The effect of fibre drawing speed on fibre elongation and fractional fibre breakage after soaking (for each fibre drawing speed, at least three fibres were analysed to calculate the elongation). Broken fibres are marked with orange arrows ($\rightarrow$), but it is of note that some broken fibres might be out of focus or field, thus, fibre breakage rate is calculated by comparing the number of fibres in dry and soaked statuses.

3.4. Fibre drawing speed and crosslinking time on controlled curling behaviours

In this section, we will establish the parametric space for tuning and controlling the fibre properties when soaked in aqueous media, as previously indicated in Fig. 2. In order to examine the suitability of the fibre devices for solution use, dry fibres produced from different drawing speeds were immersed in culture media to observe their morphological changes. Notably, fibres produced from different drawing speeds exhibit different levels of robustness for the same crosslinking concentration and time (Fig. 6a). First, the proportion of fibres damaged post-water immersion increased significantly for drawing speeds beyond 6000 mm/min, which indicates the fibres became more mechanically fragile. This could be explained by that the fibre drawing speed could affect the fibre-frame bonding region, where the ends of the suspended fibres are anchored and bonded on the PLA frames (i.e. Fig. 5b). A higher fibre drawing speed might lead to a weaker fibre-frame bonding, and this could lead to fibre breakage upon soaking in the cell culture media. Furthermore, by equating swelling-induced fibre elongation as the change in fibre length before (L_0) and after liquid soaking (L_s), with elongation = $(L_s - L_0) / L_0$, we found that an increased drawing speed led to slightly decreased swelling. For the chosen crosslinking concentration of 6 % v/v, the percentage of elongation is below $\sim 7\%$. Overall, the fibre drawing speed does not significantly affect the fibre swelling upon soaking compared to the gelatin cross-linking parameters, because the fibre swelling is mainly controlled by the degree of the gelatin crosslinking (i.e. the crosslinker concentration and crosslinking time, as discussed in the next section). Integrating the two drawing speed dependent-effects (i.e. fibre elongation and fibre breakage), Fig. 6b shows that one can deduce an optimal drawing speed, in the region of ~ 4500 mm/min, in order to yield fibre devices with visually regular, and straight fibre patterns for

aqueous solution applications.

3.5. Interplay between crosslinking concentration and time

For a specific hydrogel and crosslinker choice, the percentage of crosslinking is known to be the most important parameter controlling the hydrogel's free swelling behaviour, and hence the associated shape changes between the dry and the hydrated states. Studies using far-field electrospinning demonstrate different crosslinking strategies in affecting electrospun fibres' swelling behaviours [47]. Here, we investigate the possibility to tune and control the swelling-induced fibre curling during additive batch electrospinning patterning. In achieving this aim, the most important parameter to be investigated is the crosslinker (i.e. glyoxal) concentration. A heavily crosslinked hydrogel would be expected to have a reduced water uptake, thus a less change in volume and shape, and *vice versa*. Previously, it was found that a glyoxal concentration in gelatin of up to 10 % v/v imparted good cellular viability *in vitro* [42]. Therefore, 10 % v/v glyoxal was chosen as the upper limit concentration in our study. In addition, crosslinking time was also studied in conjunction, since it cannot be easily decoupled in our selected crosslinking strategy.

Fig. 7 shows systematic studies on the interplay between glyoxal concentration and crosslinking time and hydrated fibre morphology. Notably, although the as produced, dry fibre patterns appear similar across the different processing combinations, the solution immersed, swollen fibre patterns show different levels of curling (Fig. 7a). Increasing the concentration of the crosslinking agent drastically increased the rate of the crosslinking reaction. At the lowest concentration assessed, 3 % v/v, the fibres produced with shorter crosslinking times were wavy, noticeably thicker and more swollen as they were immersed in the solution. For the samples with longer crosslinking times, the fibres were progressively thinner and straighter. Towards the

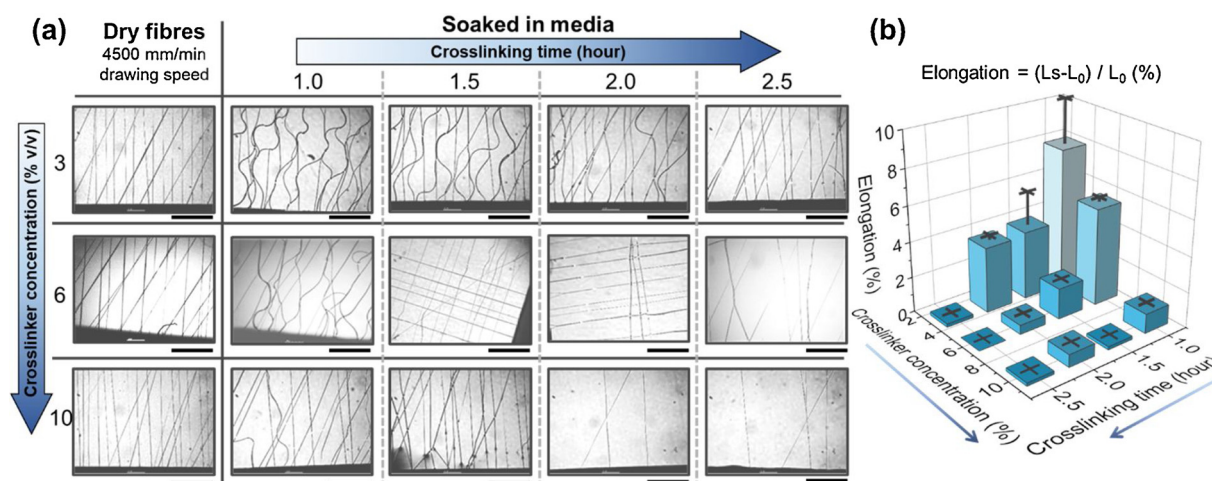


Fig. 7. (a) Images showing swelling-induced fibre curling as a result of different combinations of crosslinking time and crosslinker concentration (scale bar: 150 μm). (b) Mapping of fibre ‘Elongation $(L_s - L_0)/L_0$ ’ in the parameter space of crosslinker concentration and crosslinking time (for each situation, at least three fibres were analysed to calculate the elongation).

end of the crosslinking reaction, two hours later, the fibres remained completely straight. The intermediate concentration of 6 % v/v demonstrated a good compromise between fast crosslinking kinetics, and the ability to pattern fibres that remained straight without compromising the mechanical stability in solution. The highest concentration assessed, 10 % v/v rapidly became too viscous to draw fibres successfully but initially drew comparatively thick fibres which maintained their organisation when hydrated, appearing less swollen and elongated. Noticeably, when the crosslinking reaction was nearing completion, fibres patterned at all concentrations appear to be thinner, less swollen; and the regularity in their patterning was lost as it became increasingly difficult to initiate the fibres against the viscous resistance. Converting a measure of fibre curling shown in Fig. 7a to swelling-induced elongation $((L_s - L_0)/L_0)$, Fig. 7b shows a parametric space quantitatively guiding the selection of electrospinning parameters to design fibres of different degrees of curling.

4. Concluding remarks

We have outlined strategies to perform 3D additive batch electrospinning (3D-abES) workflow to pattern tethered, suspended hydrogel fibre devices. Design of print paths is key to enable layer-by-layer fabrication of discrete parts of multiple devices in parallel, minimising sample-to-sample variance. The as-produced dry fibres show ordered patterns as designed by digital files, where the fibre diameters lie between 2–4 μm . As the crosslinked gelatin fibres swell upon water uptake, the swollen fibre conformation can deviate from their original linear patterns of the dry state. Tethering the fibres to a rigid thermoplastic structure preserves the structural integrity of the entire fibre device. Through exploring the influence of fibre drawing speed on liquid immersed fibre integrity, an optimised fibre drawing speed of $\sim 4500 \text{ mm/min}$ was found. Exploring the processing parameter combination of crosslinking time and crosslinker concentration leads to control of the swollen fibre patterns from straight, to curly, to wavy. The tuneable patterns of the gelatin fibres could be potentially useful in mimicking the diverse morphologies of extracellular matrix fibres. The ability to produce hydrogel microfibre devices with adequate consistency may hold important applications for cell culture, bio-sensing or soft actuator applications. Furthermore, we envisage this work could contribute to making new 3D formats of suspension cultures of multicellular aggregates via guided assembly, for drug assay applications. Standardized manufacturing of the freely suspended hydrogel microfibre devices, could open possibility to use these microfibrils as an *in-situ* force probe, to evaluate tissue mechanics of multicellular

aggregates at high throughput.

Author contributions

E.L.G., W.W. and Y.Y. S. H wrote the main manuscript text. E.L.G. performed the fibre printing experiments. W.W. and R.L. prepared the figures and performed sample imaging, image analysis, and experimental repeats. Y. Y. S. H conceived and supervised the project.

CRediT authorship contribution statement

Elisabeth L. Gill: Conceptualization, Methodology, Investigation, Writing - original draft. **Wenyu Wang:** Formal analysis, Writing - original draft, Visualization. **Ruishan Liu:** Visualization, Validation. **Yan Yan Shery Huang:** Project administration, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.addma.2020.101456>.

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